

25.0.1 NIELIT 2016 DEC Scientist B (CS) - Section B: 11



A sinusoidal signal is analog signal, because:

- A. it can have a number of values between the negative and positive peaks
- B. it is negative for one half cycle
- C. it is positive for one half cycle
- D. it has positive as well as negative values

nielit2016dec-scientistb-cs non-gatecse

Answer key

25.0.2 NIELIT 2016 DEC Scientist B (CS) - Section B: 10



The noise due to random behavior of charge carriers is:

- A. Partition noise
- B. Industrial noise
- C. Shot noise
- D. Flicker noise

nielit2016dec-scientistb-cs non-gatecse

Answer key

25.0.3 NIELIT 2016 MAR Scientist B - Section C: 52



Which of the following is possible in a token passing bus network?

- A. in-service expansion.
- B. unlimited number of stations.
- C. both (A) and (B).
- D. unlimited distance.

nielit2016mar-scientistb non-gatecse

Answer key

25.0.4 NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 28



The capacity relationship is given by

- A. $C = W \log_2(1 + S/N)$
- B. $C = 2W \log_2(1 + S/N)$
- C. $C = W \log_2(1 - S/N)$
- D. $C = W \log_{10}(1 + S/N)$

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Answer key

25.1

Cryptography (1)

25.1.1 Cryptography: NIELIT 2021 Dec Scientist B - Section B: 46



Consider the following two statements:

- I. A Hash function, which is used for computing digital signatures, is an injective function
- II. An encryption technique such as DES performs a permutation on the elements of its input alphabet

- A. Both are false
 B. (I) is true (II) is false
 C. (II) is true (I) is false
 D. Both are true

nielit-2021-it-dec-scientistb cryptography hashing digital-logic

Answer key 

25.2 Data Compression (1)

25.2.1 Data Compression: NIELIT 2022 April Scientist B | Section B | Question: 97

A message is made up entirely of characters from the set $X = \{P, Q, R, S, T\}$. The table of probabilities for each of the characters is shown below:



Character	Probability
<i>P</i>	0.22
<i>Q</i>	0.34
<i>R</i>	0.17
<i>S</i>	0.19
<i>T</i>	0.08
Total	1.00

If a message of 100 characters over X is encoded using Huffman coding, then the expected length of the encoded message in bits is _____.

- A. 225 B. 226 C. 227 D. 228

nielit2022apr-scientistb probability huffman-code encoding analytical-aptitude data-compression

25.3 Information Theory (3)

25.3.1 Information Theory: NIELIT 2021 Dec Scientist B - Section B: 54

An analog signal having 3 kHz bandwidth is sampled at 1.5 times the Nyquist rate. The successive samples are statistically independent. Each sample is quantized into one of 256 equally likely levels. The information rate of the source is :



- A. 3 kbps B. 72 kbps
 C. 256 kbps D. 9 kbps

nielit2021dec-scientistb information-theory probability

25.3.2 Information Theory: NIELIT 2021 Dec Scientist B - Section B: 67

The channel capacity of a noise free channel having M symbols is given by :



- A. M B. 2^M C. $\log M$ D. None of these

